AUTOMATED TEST LOG

Institution: Institut d'Enseignement Supérieur de Ruhengeri (INES-Ruhengeri)
Course: SWE 3513 — Artificial Intelligence
Assignment: Individual Assignment 2
Project: Musanze Harvest Intelligence Service

1. Student identification

- **Full name:** MANZI PRINCE Peter
- **Registration number:** ug25/28149
- **Institutional username:** ug2528149@ines.ac.rw
- **Verification code:** AI-A2-8149
- **Random seed:** 8149

2. Testing environment

- **Operating system:** Windows, PowerShell
- **Project directory:** `C:\Users\DJO\Desktop\AI_A2_2528149`
- **Python environment:** Project virtual environment (`.venv`)
- **Test framework:** pytest
- **Application framework:** FastAPI
- **Execution date:** To be confirmed from the actual test run

3. Reproducibility pipeline execution

Command executed:

```
python run_all.py --student-id ug2528149@ines.ac.rw --seed 8149
```

Observed result: Pipeline completed successfully.

The terminal output confirmed the following:

- Student verification code: `AI-A2-8149`
- Original generated records: 271
- CSV records before cleaning: 274
- Exact duplicate rows detected: 3
- Missing soil moisture values: 14
- Missing fertilizer values: 15
- Records after duplicate removal: 271
- Training records: 216
- Test records: 55
- Selected regression model: `linear_regression`
- Selected classification model: `logistic_regression`

- Selected number of clusters: 2

Generated dataset SHA-256:

```
e3cbb00273a9802c2c57e0b3a5aa90e402ded0000a86da22a040a174d93d7d55
```

The pipeline reported that all required artifact files existed and finished successfully.

4. Required automated test coverage

The assignment requires at least six meaningful pytest tests. The following cases should be verified against the submitted implementation.

Test case	Expected behavior	Result
Valid prediction	Returns the required prediction fields	Pending verification
Missing required field	Rejects incomplete input	Pending verification
Incorrect field type	Returns a validation error	Pending verification
Out-of-range input	Rejects invalid numeric values	Pending verification
Reproducibility	Repeated runs produce the same dataset fingerprint	Pipeline run completed; repeat comparison pending
API response	Health or prediction endpoint returns the expected response	Pending verification

5. Command for automated tests

Execute the following command from the project root with the virtual environment activated:

```
pytest -q
```

Record the actual terminal output below after running the command.

Actual pytest output:

```
To be completed using the actual pytest -q output.
```

Total tests: To be recorded.

Passed: To be recorded.

Failed: To be recorded.

Skipped: To be recorded.

Overall automated test status: Pending execution and verification.

6. Additional inference and API checks

Run command-line prediction using:

```
python predict.py --json samples/valid.json
```

Check the API by starting:

```
uvicorn app:app --host 127.0.0.1 --port 8000
```

Then verify:

- `GET /health` responds correctly.
- `GET /model-info` returns the expected model and verification information.
- `POST /predict` accepts valid input and returns the required prediction fields.
- Invalid requests produce appropriate HTTP 4xx responses.

These checks must be recorded as passed only after their actual responses have been observed.

7. Conclusion

The recorded pipeline execution completed successfully, generated the individual dataset, removed duplicate rows during cleaning, trained the required machine-learning tasks, and confirmed the presence of required artifacts.

The successful pipeline execution is documented evidence of reproducibility-pipeline completion for this run. It does not, by itself, establish that all automated tests or API validation checks have passed.

The final test status must be completed using the actual pytest output and verified API responses.

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